

DAYEON (ZOEY) KI

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RESEARCH INTERESTS

- Machine Translation
- Cross-Cultural NLP
- Large Language Model Alignment
- Multilingual NLP

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science

Supervisor: Marine Carpuat

Sep. 2023 – Present

Maryland, United States

Korea University

Bachelor of Statistics, Bachelor of Computer Science and Engineering

Summa Cum Laude

Mar. 2019 – Jun. 2023

Seoul, S.Korea

International School of Prague

Feb. 2015 – May 2018

Prague, Czech Republic

PUBLICATION

(* denotes equal contribution)

[1] **GraphicBench: A Planning Benchmark for Graphic Design Generation with Language**

Dayeon Ki, Tianyi Zhou, Marine Carpuat, Gang Wu, Puneet Mathur, Viswanathan Swaminathan

Under Submission (AAAI 2026)

[2] **Should I Share this Translation? Evaluating Quality Feedback for User Reliance on Machine Translation**

Dayeon Ki, Kevin Duh, Marine Carpuat

EMNLP 2025

[3] **Beyond Benchmarks: Exploring Machine Translation Error Perception and Reliance Among the General Public**

Yimin Xiao, Yongle Zhang, Dayeon Ki, Calvin Bao, Marianna Martindale, Charlotte Vaughn, Ge Gao, Marine Carpuat

EMNLP 2025

[4] **AskQE: Question Answering as Automatic Evaluation for Machine Translation**

Dayeon Ki, Kevin Duh, Marine Carpuat

ACL 2025 Findings

[5] **Multi-LLM Agents Debate for Equitable Cultural Alignment**

Dayeon Ki, Rachel Rudinger, Tianyi Zhou, Marine Carpuat

ACL 2025 (Oral)

[6] **Automatic Input Rewriting Improves Translation with Large Language Models**

Dayeon Ki, Marine Carpuat

NAACL 2025

[7] **Guiding Large Language Models to Post-Edit Machine Translation with Error Annotations**

Dayeon Ki, Marine Carpuat

NAACL 2024 Findings

[8] **Cross-lingual QA: A Key to Unlocking In-context Cross-lingual Performance**

Sunkyoung Kim*, Dayeon Ki*, Yireun Kim, Jinsik Lee

ICML 2024 ICL Workshop

[9] Mitigating Semantic Leakage in Cross-lingual Embeddings via Orthogonality Constraint

Dayeon Ki, Hyunjoong Kim, Cheonbok Park

ACL 2024 RepL4NLP Workshop

[10] Towards Accurate Translation via Semantically Appropriate Application of Lexical Constraints

Yujin Baek*, Koanho Lee*, Dayeon Ki, Hyoung-Gyu Lee, Cheonbok Park, Jaegul Choo

ACL 2023 Findings

ACADEMIC EXPERIENCE

Visiting Scholar

Johns Hopkins University

Jun. 2025 – Aug. 2025

Baltimore, United States

- Supervisor: *Kevin Duh*
- Working on improving multilingual RAG systems and language preference phenomenon during generation.

Undergraduate Researcher

Language and Knowledge Laboratory (KAIST)

Dec. 2022 – Jun. 2023

Seoul, S.Korea

- Supervisor: *Minjoon Seo*
- Collaborated with Korea Meteorological Administration to develop weather commentary generation and weather database retrieval model.

Undergraduate Researcher

Data and Visual Analytics Laboratory (KAIST)

Feb. 2022 – Nov. 2022

Seoul, S.Korea

- Supervisor: *Jaegul Choo*
- Collaborated with NAVER Papago to develop context-aware neural machine translation model in lexically constrained settings [6].

INDUSTRY EXPERIENCE

Research Scientist Intern

Adobe Research

May. 2024 – Dec. 2024

California, United States

- Developed a multi-agent planning benchmark 'CreativeTown' for creative content generation tasks.
- Evaluated performance of multimodal LLM agents on both planning and execution stage.

Research Intern

NAVER Cloud, Papago

Jan. 2023 – Jun. 2023

Seoul, S.Korea

- Focused on solving semantic leakage, where a significant amount of language-specific information is unintentionally leaked into semantic representations, present in current representation learning methods.
- Developed model with our training objective, Orthogonality Constraint Learning (ORACLE) to disentangle language bias and semantics in cross-lingual sentence embeddings, leading to effective parallel data mining [7].

Research Intern

LG AI Research

Aug. 2022 – Dec. 2022

Seoul, S.Korea

- Participated in developing and evaluating LG's billion-scale bilingual Large Language Model, EXAONE.
- Experimented shot variations and few shot example selection methods leveraging re-ranking models based on similarity metrics, which led to improved generalization ability than previous methods.
- Developed novel cross-lingual prompting techniques to enhance cross-lingual transfer of Large Language Models [6].

Co-Founder & Software Engineer

Da-Chae

Aug. 2020 – Dec. 2021

Seoul, S.Korea

- Developed convolutional neural network-based framework which updates parameters in the direction of minimizing parameters between image pixels.
- Enhanced UI/UX flow for better recommendation of paintings based on users' interior theme.
- Participated in multiple investment meetings with Donghwa Corp. and Hankook Ilbo.

TEACHING

(CMSC320) Introduction to Data Science <i>Teaching Assistant</i>	Aug. 2023 – Dec. 2023
(GEST151) Data Science and Artificial Intelligence <i>Student Mentor</i>	Mar. 2022 – Jun. 2022
(GEST151) Data Science and Artificial Intelligence <i>Student Mentor</i>	Mar. 2021 – Jun. 2021
• Nominated as <i>Best Mentor</i> through mentors and students' survey. • Instructed weekly lectures and lab sessions using R and Python. • Guided 60+ students' final projects in preparation for the Data Science Grand Conference.	
KUBIG NLP BoostCamp Course <i>Lecturer</i>	Jan. 2021 – Feb. 2021

EXTRACURRICULAR ACTIVITIES

Vizable <i>Team-Leader, Staff</i>	Jan. 2022 – Jun. 2022
• Studied and created various dashboards using data visualization tool Tableau.	
KUBIG (Data Science Society of Korea University) <i>President</i>	Dec. 2020 – Jul. 2022
• Led multiple collaboration meetings with IT service companies (<i>Makina-Rocks</i>) for industry-academic research collaboration projects. • Conducted team-based projects: Lecture review summarization, GPT-2 based Curriculum Vitae generation. • Instructed Winter 2021 NLP Bootcamp course for the whole society members.	
KUBA (Korea University Buddy Assistant) <i>Staff</i>	Jul. 2020 – Dec. 2020
• Planned and organized peer mentoring programs for the international exchange students at Korea University.	
Koala Univ. Data Science Club <i>Member</i>	Mar. 2020 – Jun. 2020
• Lead multiple courses on data crawling, data processing and machine learning based analyzing techniques. • Participated and advanced as finalist at <i>Koala Univ Hackathon</i>.	

INVITED TALKS

Oct. 2024	University of Toronto, Ontario Tech University
Oct. 2017	TEDxYouth ISP
Mar. 2017	Applied Linguistics Presentation @ Charles University

HONORS & REWARDS

2023-2025	Dean's Scholarship, Department of Computer Science, University of Maryland, College Park
Dec. 2022	Dean's List Scholarship, Department of Statistics, Korea University
Aug. 2021	Finalist, Korea University Capstone Datathon
Jun. 2021	Best Teaching Mentor, Data Science and Artificial Intelligence
Apr. 2021	Tableau Desktop Specialist
Apr. 2021	SQLD (SQL Developer), Korea Data Agency
Apr. 2021	ADSP (Advanced Data Analytics semi-Professional), Korea Data Agency
Oct. 2020	Runner-up, Seoul Entrepreneurship Competition
Aug. 2020	Runner-up, Korea Artificial Intelligence Hackathon
2016-2017	University of Waterloo Math Representative

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Developer Tools: Docker, VS Code, PyCharm

Technologies/Frameworks: Pytorch, Tensorflow, Linux, Github, LaTeX, HuggingFace, Tableau

LANGUAGE PROFICIENCY

Bilingual in English and Korean (native)

- International school experience in China (2004-2011) and Czech Republic (2015-2018)
- TOEFL iBT : 111/120 (Reading: 29, Listening: 30, Speaking: 27, Writing: 25)
- SAT : 1460/1600 (Reading & Writing: 660, Math: 800)

Conversational in French

- DELF B2

Beginner in Chinese